SWIFT SHIP TRACKER

Book. Track. Deliver. Smarter.

User Story

The SwiftShip Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. Introduction

1.1 Project Overview

The SwiftShip Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose

The purpose of SwiftShip Tracker is to:

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.
- Improve agent efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access to customers, agents, support staff, and administrators.

2. Ideation Phase

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	book a parcel easily	booking requires support assistance or multiple applications	parcel operations are not centralized	frustrated
PS-2	Customer / Sender	track my parcel	I do not have a simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	update parcel status	status updates require manual handling	there is a need for centralized operational tracking	concerned
PS-4	Customer	receive delivery updates	notifications may not be available at every delivery stage	communication needs to be automated	uncertain
PS-5	Admin	monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

2.2 Empathy Map Canvas

User Persona: Customer / Sender

Think & Feel
- Wants a simple way to book and track parcels without repeatedly contacting support.
- Hopes the parcel reaches on time and safely.
- Feels more confident with real-time updates.
- Gets concerned when there is no tracking information.

Hear
- Delivery confirmations and updates from the parcel service.
- Information about delivery timelines.
- Feedback from friends, family, or colleagues about delivery experiences.
- Announcements or notifications about service changes.

See
- Parcel status and tracking information.
- Delivery details and estimated delivery date.
- User-friendly portal or mobile interface.
- Notifications and emails about shipment progress.



Say & Do
- Books parcels through the portal or app.
- Checks shipment status regularly.
- Asks for delivery updates when needed.
- Provides feedback on delivery experience.
- Shares tracking information with recipients.

Pain
- Delays in delivery.
- Confusion due to lack of clear tracking information.
- Difficulty accessing shipment details.
- Dependence on customer support or multiple apps.
- Uncertainty about delivery timelines.

Gain
- Easy and convenient parcel booking.
- Real-time tracking and delivery updates.
- Timely notifications at every stage.
- Better visibility of delivery progress.
- Peace of mind knowing the parcel is on track.

Figure 1. Empathy Map Canvas - Customer / Sender

2.3 Brainstorming & Idea Prioritization

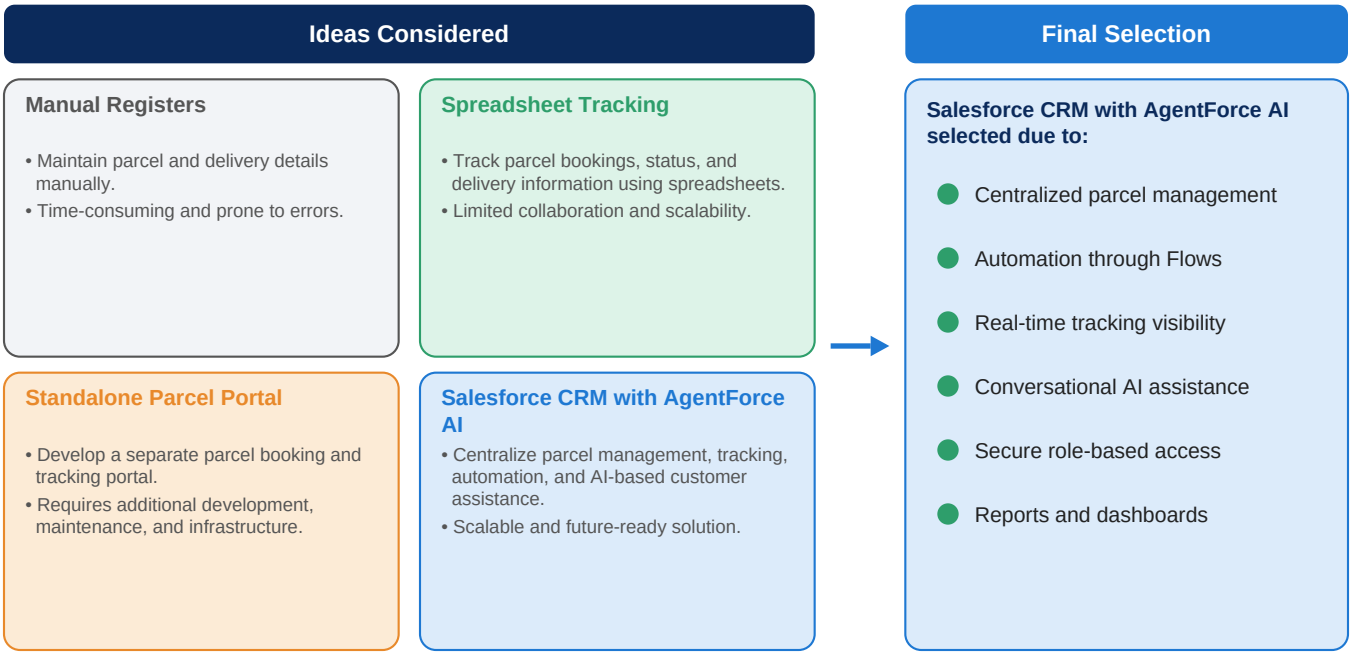


Figure 2. Ideas considered and final selection

3. Requirement Analysis Phase

3.1 Customer Journey Map

Stages: Parcel Booking, Parcel Record Creation, Dispatch & Shipment, In-Transit Tracking, Out for Delivery, Delivery Confirmation, Customer Notification, and Feedback / Support.

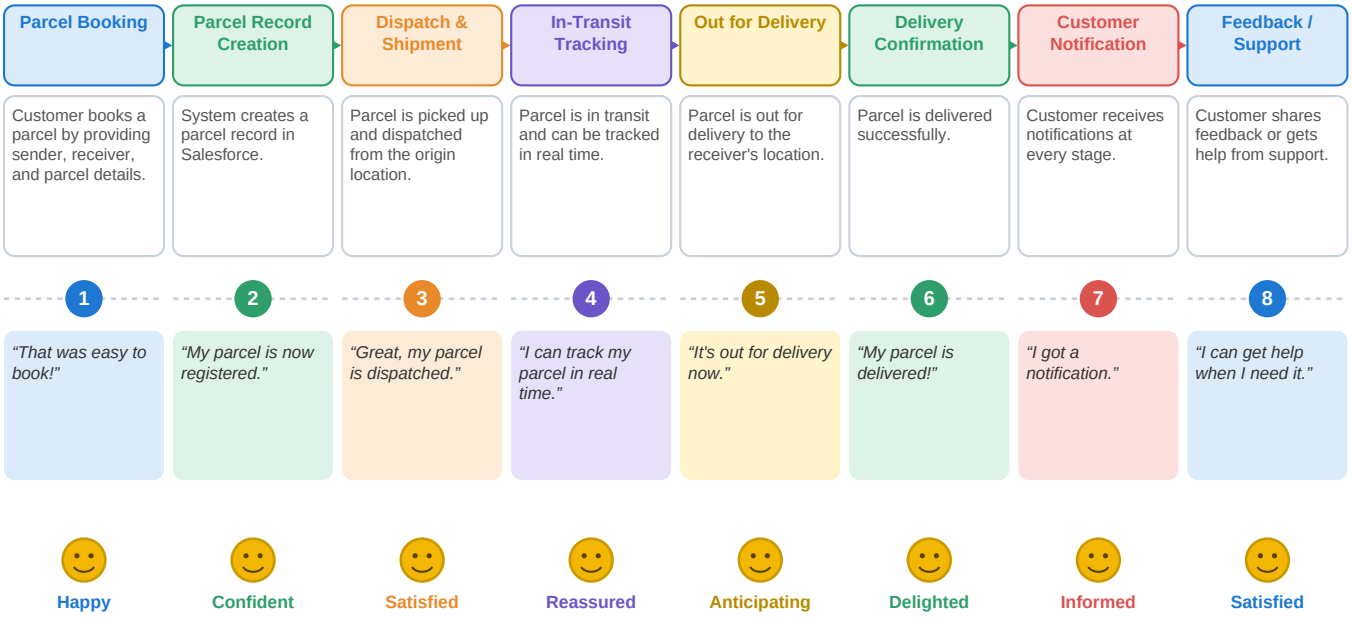


Figure 3. Customer Journey Map - from booking to delivery (top: customer thoughts, bottom: customer emotion)

3.2 Solution Requirements

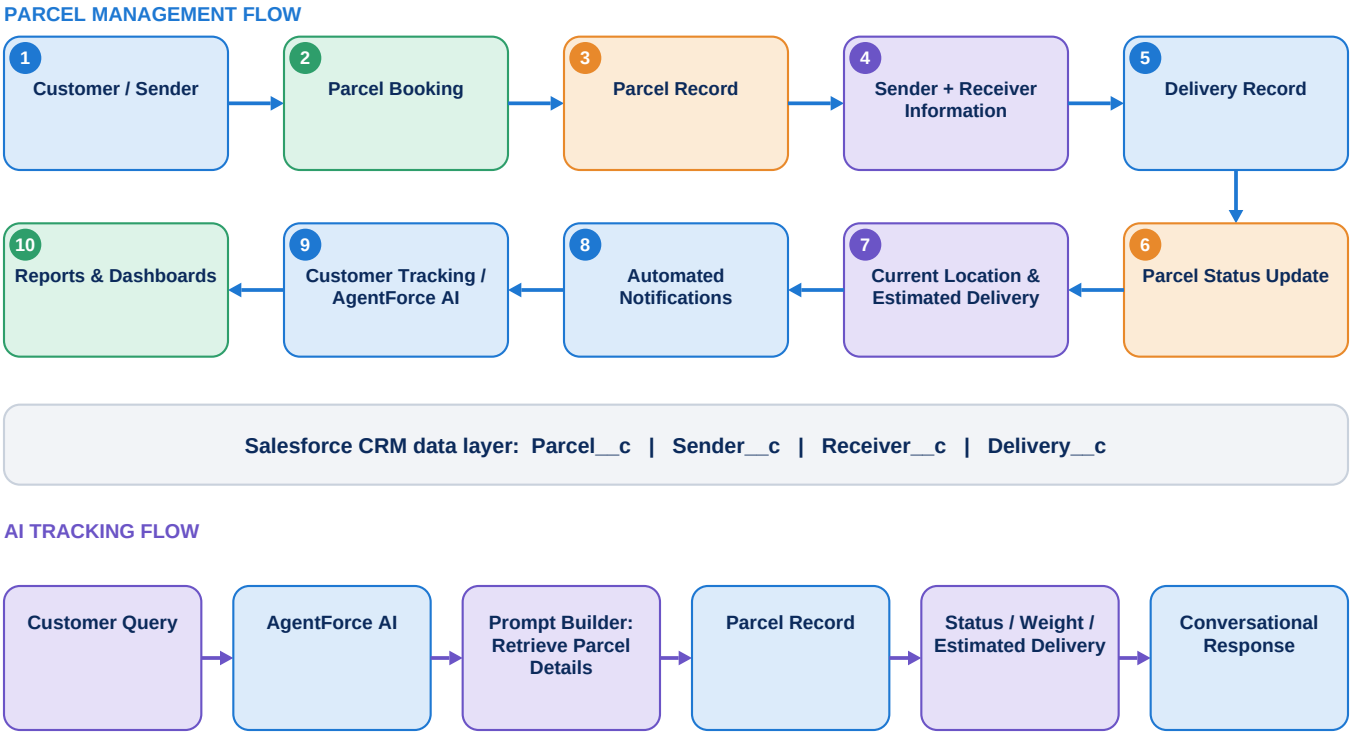
Functional Requirements

FR No	Requirement
FR-1	Parcel booking and record creation
FR-2	Parcel status tracking
FR-3	Weight management
FR-4	Estimated delivery date management
FR-5	Sender information management
FR-6	Receiver information management
FR-7	Delivery and current-location tracking
FR-8	Agent parcel status updates
FR-9	Automated customer notifications
FR-10	AI-powered parcel tracking
FR-11	Reports and dashboards
FR-12	Customer access through Experience Cloud

Non-Functional Requirements

NFR	Description
Usability	Intuitive Salesforce and customer-facing interface
Security	Role-based and field-level access control
Reliability	Accurate parcel status and delivery information
Performance	Timely Flow and AgentForce responses
Availability	Salesforce cloud-based access
Scalability	Supports growing parcel and delivery operations
Data Integrity	Controlled access and relationship-based parcel data
Maintainability	Flows, prompts, topics, and configurations can be updated

3.3 Data Flow Diagram



The Prompt Builder template retrieves Parcel ID, Status, Weight, and Estimated Delivery Date for tracking responses.

Figure 4. End-to-end parcel management flow and AI tracking flow

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience Cloud
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates
Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem-Solution Fit

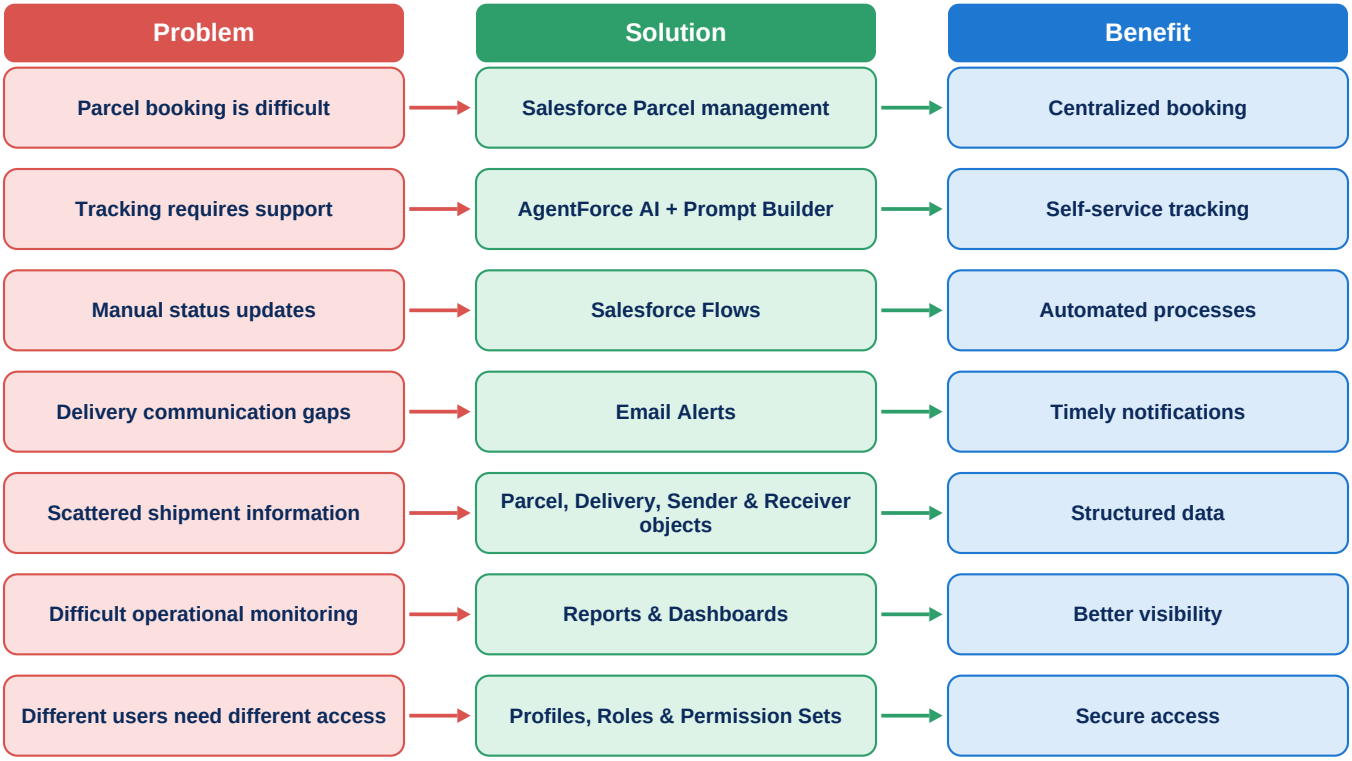


Figure 5. Problem-Solution Fit

4.2 Proposed Solution

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

4.3 Solution Architecture

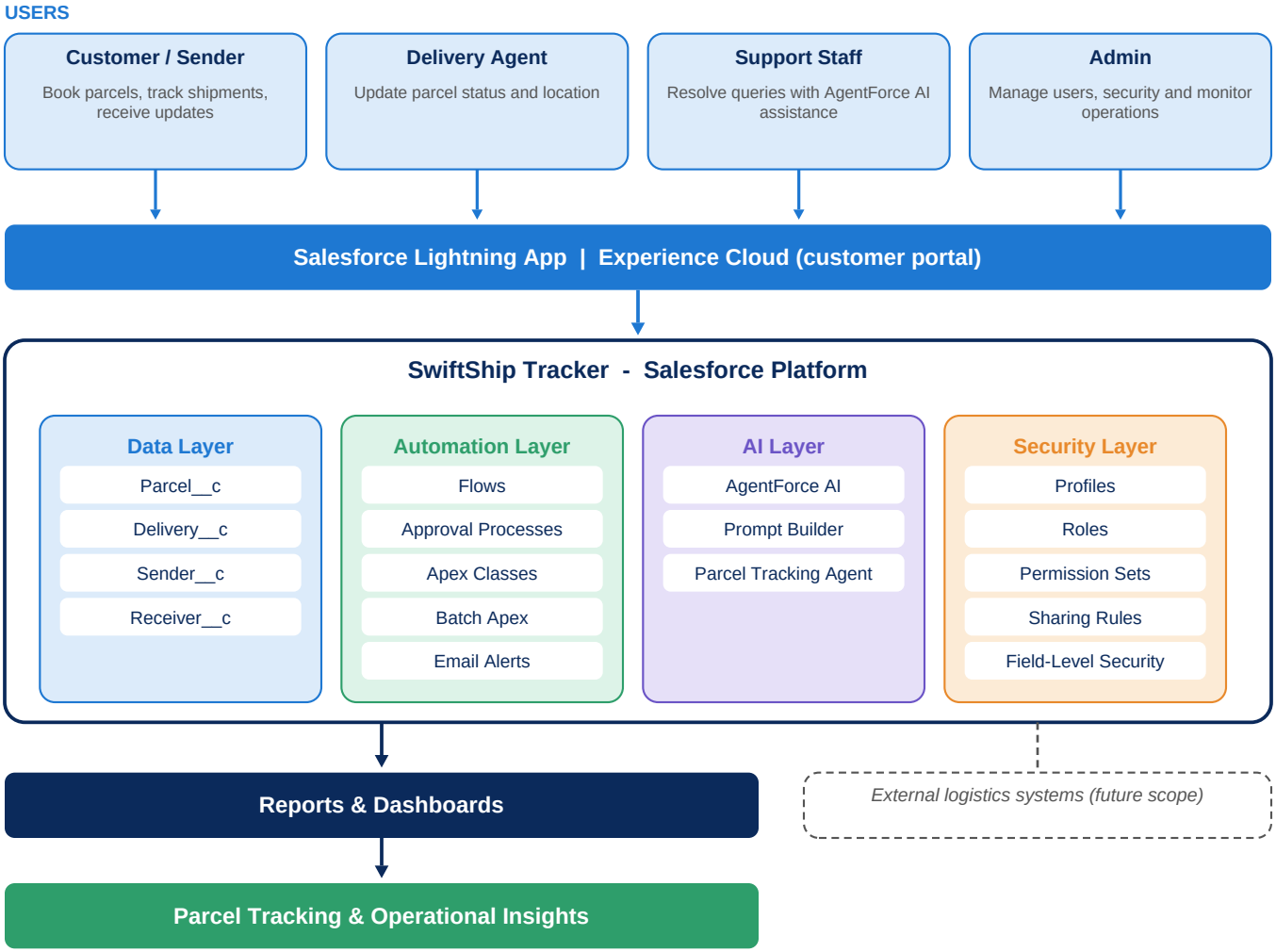


Figure 6. SwiftShip Tracker solution architecture

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics, Stories, Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Member
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShip Tracker parcel management system.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender, and Receiver objects with proper relationships so that parcel and delivery data is stored and managed systematically.	5	High	Member2

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Member
Sprint-2	Data Modeling	USN-3	As a staff member, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily navigate and manage parcel and delivery operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate parcel details, mandatory fields, and booking inputs so that invalid or incomplete parcel records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a delivery coordinator, I want automated Flows to update parcel status and delivery information so that parcel processing requires minimal manual effort.	3	High	Member4
Sprint-4	Apex	USN-6	As a system, I want Apex triggers and classes to automatically perform critical parcel and delivery actions so that important processes are handled accurately.	5	High	Member1
Sprint-4	Apex	USN-7	As an operations manager, I want Batch Apex jobs to process and update parcel or delivery records automatically so that operational data remains accurate and up to date.	5	High	Member2
Sprint-5	Reports	USN-8	As a manager, I want reports showing parcel status, delivery performance, and operational metrics so that I can monitor SwiftShip operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards & AI	USN-9	As a manager, I want dashboards showing parcel status, delivery performance, and key metrics, while Agentforce provides AI-powered tracking assistance, so that I can monitor operations efficiently.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		
Sprint-5	20	6 Days	21 Nov 2022	26 Nov 2022		
Sprint-6	20	6 Days	28 Nov 2022	03 Dec 2022		

6. Project Development Phase

Phase 1: Requirement Analysis & Planning

Objective

“Customers often face delays and confusion while booking, tracking, and managing parcel deliveries because they need to call support or use multiple apps. A conversational AI agent can simplify this by allowing customers to book parcels, get live tracking, and manage deliveries in one place.”

Key Requirements

- Requirement gathering from end users and delivery agents.
- Defining system scope and user personas (Customer, Agent, Admin).
- Designing the data model with relationships between Parcel, Sender, Receiver, and Delivery.

- Identifying use cases such as parcel booking, tracking, and delivery update automation.
- Selecting tools: Salesforce Flows, Prompt Builder, and AgentForce AI.

Defining Project Scope & Objectives

Project Scope

- Build a scalable CRM solution for parcel delivery businesses to manage customer bookings, shipments, and delivery operations efficiently.
- Support the end-to-end parcel lifecycle: parcel booking, dispatch, tracking, and delivery confirmation.
- Include comprehensive data model design, automation (Flows, Apex, Email Alerts), AI-powered Prompt Builder, and reporting dashboards for effective management.
- Provide role-based access for admins, delivery agents, and customers through Salesforce profiles, permission sets, and Experience Cloud.
- Ensure real-time parcel visibility for customers and simplified operational tracking for agents through AI-integrated conversational support.

Objectives Summary

- Streamline parcel operations by automating booking, tracking, and delivery notifications to reduce manual updates.
- Improve customer satisfaction with transparent, real-time shipment status and self-service tracking options.
- Enhance agent efficiency through centralized dashboards and guided Flows for quick status updates.
- Enable business insights with reports and dashboards showing parcel trends, delivery performance, and agent activity.
- Ensure security and reliability by implementing Salesforce role hierarchies, sharing rules, and audit-based data access.

Gathering & Analysing User Needs

Users Involved

- **Customers / Senders** - End users who book parcels, track deliveries, and receive real-time updates through the portal or AI agent.
- **Delivery Agents** - Manage assigned parcels, update delivery status, and ensure timely handover to customers.
- **Customer Support Team** - Handle delivery-related issues, respond to customer queries, and ensure smooth communication.
- **Admin Team** - Configure the Salesforce environment, manage user roles, security settings, and monitor operational performance.
- **System Integrator / Developer** - Responsible for implementing automation, AI agent setup, and backend integrations.

Key Functional Needs

- Seamless parcel booking and tracking interface for customers through Salesforce Experience Site or agent assistance.
- Ability for agents to update parcel status (Booked, In Transit, Out for Delivery, Delivered) in real time.
- Automated notifications and email alerts to customers on every delivery stage update.
- Support for Sender and Receiver details management with address, contact, and email fields for communication.
- AI Agent integration (via Prompt Builder) for real-time parcel status queries and automated assistance.
- Reports and dashboards to monitor parcel flow, delivery performance, and agent activity for admins.
- Security controls with role-based access, profiles, and permission sets for different user levels.

Tools Used

- Salesforce Developer Org - for configuration, development, and testing.
- Prompt Builder (AgentForce AI) - to create intelligent parcel tracking prompts and responses.

- Salesforce Flow Builder - to automate booking, tracking, and delivery updates.
- Google Sheets / Docs - for initial data planning, requirement documentation, and system mapping.

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned

- **Custom Objects:** Parcel__c, Delivery__c, Sender__c, Receiver__c
- **Standard Objects:** Account, Contact, Case, Task, EmailMessage, User
- **Automations:** Record-Triggered Flows, Auto-Launched Flows, Approval Processes, Scheduled Flows
- **Apex:** Apex Classes for automated parcel operations (e.g., parcel status updates, delivery notifications), Batch Apex for periodic updates or notifications
- **UI:** Lightning App Page (SwiftShip Tracker App), Dynamic Forms for Parcel Details, Lightning Record Pages, and Experience Cloud for Customer Access
- **AgentForce AI & Prompt Builder:** Integrated AI agent trained to fetch parcel data based on Parcel ID and respond to tracking or delivery status queries
- **Email Services:** Custom Email Alerts and Templates for booking confirmations, delivery updates, and completion acknowledgments
- **Security:** Profiles, Permission Sets, Role Hierarchies, Field-Level Security, and Sharing Rules to ensure controlled access for Admin, Agent, and Customer roles

Designing Data Model and Security Model

Data Model Includes

Object	Purpose and relationships
Parcel__c	Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date. Linked to Sender__c and Receiver__c objects to fetch sender and receiver details. Connected with Delivery__c object to track real-time parcel location and status.
Delivery__c	Stores delivery-specific information such as current location, estimated delivery date, and linked parcel. Connected with Sender__c and Parcel__c for relationship-based tracking. Used by agents to update and monitor delivery status during the parcel lifecycle.
Sender__c	Captures sender-related information such as address, contact number, and email. Linked to multiple parcels to track sender history and associated deliveries. Used for sending booking confirmations and delivery updates via automated emails.
Receiver__c	Stores receiver details such as address, contact, and email for communication and delivery confirmation. Linked to Sender__c and Parcel__c to associate both ends of the shipment process. Used by agents for delivery reference and automated email notifications upon completion.

Security Model Design

- **Role Hierarchy:** Administrator, Delivery Manager, Agent, Customer
- **Profiles:** Admin Profile - full access to all objects and records. Delivery Agent Profile - Read/Edit access to Parcel, Delivery, Sender, and Receiver objects. Customer Profile - read-only access to their own parcel details via Experience Cloud. Support Staff Profile - access to Cases and Parcel details for issue resolution.
- **Permission Sets:** Used to grant additional privileges (e.g., “Update Delivery Status,” “Access Agent Console”).
- **Record-Level Security:** Sharing Rules applied for Parcel__c and Delivery__c to allow controlled access for agents and managers. Customers can only view records linked to their own Sender or Receiver details.
- **Field-Level Security:** Sensitive fields like Sender Email, Receiver Contact, and Location Coordinates restricted to Admin and Agent profiles only.